

Python Logging Module Reference: Basic Usage

The logging module provides a flexible framework for emitting log messages from Python programs. The five standard levels, in increasing severity, are DEBUG, INFO, WARNING, ERROR, and CRITICAL. Use `logging.basicConfig()` to configure the root logger's level and output format before any log calls are made elsewhere in the program.

Python Logging Module Reference: Handlers and Formatters

Handlers determine where log records go: `StreamHandler` sends them to console, `FileHandler` writes them to a file, and `RotatingFileHandler` rotates log files once they reach a configured size. Each handler can have its own `Formatter`, allowing different destinations to show different levels of detail from the same log call.

Python Logging Module Reference: Best Practices

Avoid using the root logger directly in library code; instead call `logging.getLogger(__name__)` so consumers of your library can control verbosity independently. Never log sensitive data such as passwords or API keys. Use lazy formatting, e.g. `logger.info("value: %s", x)`, instead of f-strings, to avoid the formatting cost when the log level suppresses it.